```
#include<stdio.h>
#include<malloc.h>
struct node
{
	int data;
	struct node *prev,*next;
};
struct node *head,*newnode,*temp,*tail,*before,*curr;
int total_elements=0;

void delete(int data)
{
	int found = 0;
	before = NULL;
	curr = head;
	while(curr != NULL)
	{
		if(curr->data == data)
		{
			found=1; break;
		}
		before = curr;
		curr = curr->next;
	}
	if(found == 0)
	{
		printf("\n element not found");
	}
	else
	{
		if(curr == head)
		{
		  head = curr->next;
		  head->prev = NULL;
		  free(curr);

		}else if(curr->next == NULL)
		{

			before->next = NULL;
			tail = before;
			free(curr);
```

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                }else{

                        before->next = curr->next;
                        curr->next->prev = before;
                        free(curr);
                }
                total_elements--;
        }
}
void insert(int data,int pos)
{
        if(pos<=0 || pos>total_elements)
        {
                printf("\n invalid position");
                return;
        }

        int i=1;
        before = NULL;
        curr = head;
        while(temp != NULL)
        {
                if(i == pos)
                {
                        break;
                }
                before = curr;
                curr = curr->next;
                i++;
        }
}
void prepend(int data)
{
        newnode = (struct node*)malloc(sizeof(struct node));
        newnode->data = data;
        newnode->next = newnode->prev = NULL;

   if(head == NULL)
   {
      head = newnode;
        }
        else
        {
```

```
                newnode->next = head;
                head->prev = newnode;
                head = newnode;
        }
        total_elements++;
}
void append(int data)
{
        newnode = (struct node*)malloc(sizeof(struct node));
        newnode->data = data;
        newnode->next = newnode->prev = NULL;

        if(head == NULL)
        {
                head = newnode;
        }
        else
        {
                temp = head;
                while(temp->next != NULL)
                {
                        temp = temp->next;
                }

                temp->next = newnode;
                newnode->prev = temp;

                tail = newnode;
        }
        total_elements++;
}
void display()
{
                temp = head;
                while(temp != NULL)
                {
                        printf("%d->",temp->data);
                        temp = temp->next;
                }

}
void reverse()
{
```

```
                temp = tail;
                while(temp != NULL)
                {
                        printf("%d->",temp->data);
                        temp = temp->prev;
                }
}


int main()
{
        addend(10);
        addend(20);
        addend(30);

   display();
   printf("\n total elements:%d",total_elements);
        printf("\n----\n");
        delete(30);
   display();
        printf("\n total elements:%d",total_elements);
  //reverse();

}
```